Session 4 (BIAS & EFFECT): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · *#ClearDataClearImpact* · Week 4 · Saturday 21 November 2026 · online, 19:30–22:30

Name: _____ **Department:** _____

Type your answers in the Word file, or print the PDF. Have a pen, paper and a calculator (or your phone) ready. Simple division is enough.

Part	Task	When
A	Name the main bias in 4 scenarios, and one way to reduce it	End of section 4.1
B	Confounding or effect modification? (3 scenarios)	At home (optional)
C	Two 2x2 calculations	After the break
D	PPV in two settings	At home (optional)
E	Read a forest plot and choose the risk-of-bias tool	End of section 4.3
F	Proposal builder 4: draft sections 2.4 to 2.6 of your proposal	In class, end of the session. Anything left: at home

Parts C and E also have **extra practice** (self-study) for after the session.

Our running question: in adults with septic shock, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?

Abbreviations: ICU = intensive care unit, CI = confidence interval, RCT = randomised controlled trial, RR = risk ratio, RD = risk difference, NNT = number needed to treat, OR = odds ratio, PPV and NPV = positive and negative predictive value.

Part A: Spot the bias

Task. For each scenario, write the main bias (choose from the list) and one practical way to reduce it. You will vote for the bias in a poll, so have your answers ready.

> **Choose from:** selection bias · recall bias · observer bias · confounding

#	Scenario	Main bias	One way to reduce it
1	In a case-control study, families of patients who died of sepsis remember the days before ICU admission better than families of survivors.		
2	In an unblinded trial of restrictive fluids, the doctor who knows which treatment each group received decides when the patient is "ready for ICU discharge".		
3	A study of fluid balance and death includes only patients who stayed in the ICU for 3 days or more.		
4	In a registry, patients with a higher lactate received more fluid and also died more often. The authors conclude that fluid increases mortality.		

Part B: Confounding or effect modification? (homework, optional)

Task. Each study splits its result by a third factor. Compare the strata **with each other** and **with the overall (crude) result**. Write your answer and what the authors should report.

> **Choose from:** confounding · effect modification (interaction) · neither
>
> Rule of thumb: strata similar to each other but different from the crude result mean confounding. Strata different from each other mean effect modification.

#	Study (hypothetical)	Overall RR	Stratum 1	Stratum 2	Your answer	What to report
1	Vasopressor X vs Y in septic shock and death by day 28. Clinicians chose X mainly for patients with a high lactate.	1.80	lactate below 4: RR 1.00	lactate 4 or more: RR 1.00		
2	Restrictive vs usual fluids and death by day 28	0.70	age below 65: RR 0.50	age 65 or over: RR 0.95		
3	A nurse-led charting protocol and incomplete fluid charts	0.80	day shift: RR 0.80	night shift: RR 0.79		

Part C: Two 2x2 calculations

Task. Work alone on both calculations. For each one, fill in the 2x2 table, answer the questions and write the result in **one sentence** a colleague without research training would understand. Be ready to share your answers when the instructor asks.

- Draw the table this way: exposure (or treatment) in **rows**, with the exposed or treated group on top. Outcome in **columns**, with "outcome yes" on the left.
- Round to two decimals.

Reminder

Measure	Formula (from the table a, b, c, d)
Risk in exposed or treated	a / (a + b)
Risk in unexposed or control	c / (c + d)
Risk ratio (RR)	risk exposed ÷ risk control
Risk difference (RD)	risk exposed minus risk control
Number needed to treat (NNT)	1 ÷ |RD| (RD written as a proportion, e.g. 0.05)
Odds ratio (OR)	(a × d) ÷ (b × c)

Calculation 1: a fluid trial

A hypothetical trial randomised 400 adults with septic shock to a restrictive fluid strategy (200 patients) or usual care (200 patients). By day 28, **20** patients had died in the restrictive group and **30** in the usual-care group.

	Died by day 28: YES	Died by day 28: NO	Total
Restrictive	a =	b =	
Usual care	c =	d =	

1. Risk with restrictive fluids = _____ Risk with usual care = _____
2. RR = _____ 3. RD = _____ 4. NNT = _____ 5. OR = _____
6. The worked example in the slides had the same RR (20/100 vs 30/100). Why is the NNT different here? _____
7. One sentence for a colleague: _____

Calculation 2: a fluid case-control study

Investigators selected **40** ICU patients with sepsis who needed new dialysis (cases) and **80** ICU patients with sepsis from the same units who did not (controls). They looked back at each patient's fluid balance on day 3. Among the cases, **24** had a positive balance and 16 had not. Among the controls, **16** had a positive balance and 64 had not.

	Cases (new dialysis)	Controls (no dialysis)
Positive fluid balance on day 3	a =	b =
Zero or negative balance	c =	d =

1. OR = _____

2. Why can you **not** calculate the risk of dialysis, the RR or the NNT from this study?
   _____
3. Could the arrow run the other way here (reverse causation)? Why?
   _____
4. One sentence for a colleague: _____

Note: in a case-control study the columns are the case and control groups. The OR formula (a × d) ÷ (b × c) stays the same.

Extra practice (self-study): another topic

A trial randomised 400 adults on a medical ward to a new pressure-relieving mattress (200 patients) or a standard mattress (200 patients). A new pressure injury developed in **10** patients on the new mattress and in **20** on the standard mattress. Draw the table and calculate the two risks, RR, RD, NNT and OR. Is the OR close to the RR here? Why?

Part D: PPV in two settings (homework, optional)

Task. The same test has **sensitivity 90%** and **specificity 80%**. Complete each table for 500 people, then calculate PPV and NPV and answer the questions.

> Disease present = prevalence × 500. True positives (TP) = sensitivity × disease present. True negatives (TN) = specificity × disease absent. FP = false positives, FN = false negatives.

Setting 1: ICU, prevalence 20%

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = TP / (TP + FP) = _____ NPV = TN / (TN + FN) = _____

Setting 2: screening, prevalence 2%

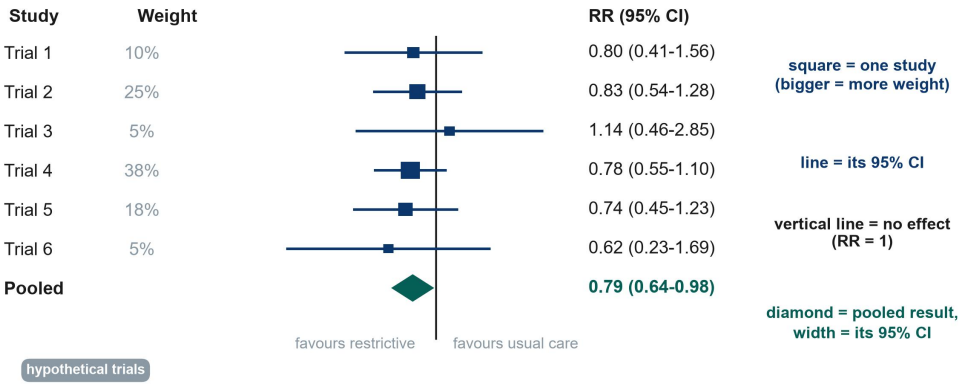
	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = _____ NPV = _____

1. Why is the PPV so different when the test is the same? _____
2. Calculate the positive likelihood ratio, LR+ = sensitivity / (1 − specificity) = _____. Does it change between the settings? _____
3. Optional: repeat for a prevalence of 50%.

Part E: Read a forest plot and choose the risk-of-bias tool

Task. Work alone. Answer questions 1 to 3 about the forest plot, then choose the risk-of-bias tool for studies a to c. Be ready to share your answers when the instructor asks. The plot is the one shown in section 4.3: six hypothetical trials of restrictive vs usual fluids, outcome death, fixed-effect pooling.



Trial	Restrictive: died / total	Usual care: died / total
Trial 1	12 / 60	15 / 60
Trial 2	30 / 150	36 / 150
Trial 3	8 / 40	7 / 40
Trial 4	45 / 250	58 / 250
Trial 5	20 / 100	27 / 100
Trial 6	5 / 30	8 / 30

Reading the plot

1. Which trial carries the most weight? Why? _____
2. Does any single trial show a clear effect (a 95% CI that does not cross 1)?
   _____
3. Does the pooled diamond cross RR = 1? Write the pooled result in one plain sentence.
   _____
4. (Self-study) Trial 3 points the other way (RR 1.14). Should it worry you? Why or why not?
   _____

Choosing the tool. Write one tool for each study.

> **Choose from: RoB 2** (Cochrane risk-of-bias tool for randomised trials) · **ROBINS-I** (non-randomised studies of interventions) · **Newcastle-Ottawa Scale (or ROBINS-E)** (cohort or case-control studies of exposures) · **QUADAS-2** (diagnostic accuracy studies) · **AMSTAR 2** (systematic reviews)

#	Study	Tool
a	Paper 1, Boulet 2024: randomised trial of restrictive vs usual fluids	
b	Paper 2, Hyun 2023: registry cohort, positive vs negative fluid balance	
c	Systematic review and meta-analysis of fluid-restriction trials	
d	(Self-study) Before-after study of a fluid protocol on one unit	
e	(Self-study) Point-of-care lactate compared with the laboratory reference test	
f	(Self-study) Cluster-randomised trial of a fluid-charting protocol (12 ICUs)	

Part F: Proposal builder 4 (exposure, outcomes and biases)

Task. Draft sections 2.4 to 2.6 of your proposal for your own question. First watch the instructor fill them in for the running example, then write your own in class (about 16 minutes). If you do not finish, complete the rest at home. Then copy your answers into Part 2 of the proposal template (`Research_Proposal_Template.docx`). Sections 2.1 to 2.3 (design, setting, population) were drafted in Session 3. For every outcome write **what**, **how** and **when**.

My research question: _____

Section	What to write	My answer
2.4 Intervention or exposure, and comparator	What exactly is the exposure or intervention, and how is it defined or measured? Who is the comparison group?	
2.5 Outcomes	**ONE primary outcome**: what it is, how and when it is measured. **2-3 secondary outcomes.**	
2.6 Main threats of bias and how they are reduced	Name at least **two** threats (selection, information, confounding) and the step in your design that reduces each.	

My claim at the end will be: ☐ "caused" (randomised design) ☐ "associated with" (observational design), because _____

Self-check before you finish

- ☐ There is a comparison group, or I explain why my study is descriptive.
- ☐ There is only ONE primary outcome, with a method and a time point.
- ☐ If my design is observational, I have named the main confounders (think: confounding by indication).
- ☐ Each bias has a concrete mitigation step ("we will be careful" is not a mitigation).
- ☐ Part 2 is complete: sections 2.1 to 2.3 (Session 3) and 2.4 to 2.6 (today).